

Radio and Far-IR Emission Associated with a Massive Star-forming Galaxy Candidate at $z \simeq 6.8$: A Radio-Loud AGN in the Reionization Era?

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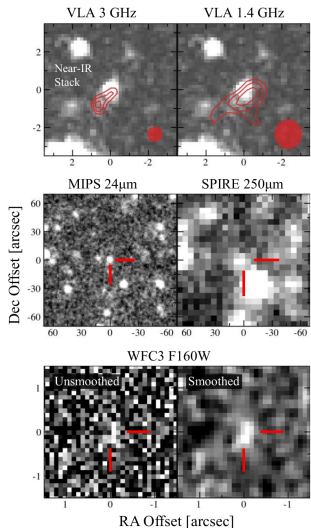
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Introduction

- Some of the early massive, UV-luminous galaxies may have carved out large ionised regions in neutral IGM during the universe reionization
- Large samples of UV-luminous reionization-era systems have now been identified by wide-area, ground-based fields
- An essential next step is to better understand the characteristics of the most massive reionization-era galaxies, including those which are substantially reddened by dust
- Additionally deliver important clues into the properties of early supermassive black holes

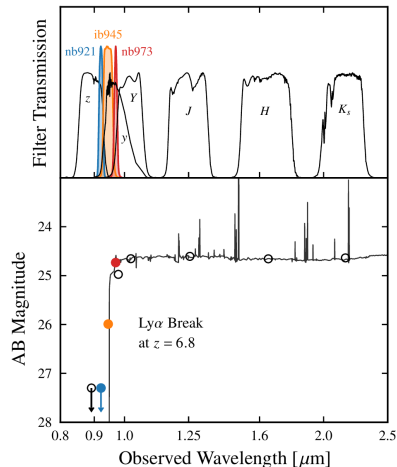
Observations of COS-87259



- [Endsley et al. 2021a](#) (hereafter E21) selection: $41\ z \simeq 6.6 - 6.9$
UV-bright, star-forming galaxies from COSMOS. **COS-87259** is one of them.
- Only source in the [E21](#) sample which has radio, sub-mm, infrared, optical observations.
- One of the most massive and oldest sources in the sample.
- Significant UV-optical reddening from dust.

Observations of COS-87259

- Reionization at $z \simeq 7$: $\text{Ly}\alpha$ break lies at $0.9 \sim 1 \mu\text{m}$
- Deep imaging with HSC
nb921, ib945 and nb973 acts as a low-resolution spectra



Observations of COS-87259

Optical to Near-Infrared

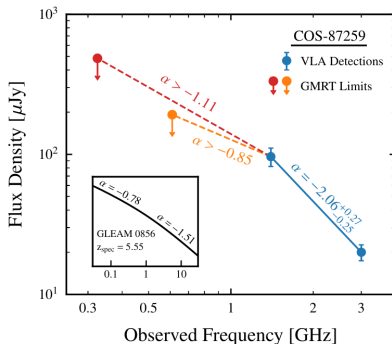
- Undetected in all bands blueward of nb973 (9730 Å)
- Significantly higher flux on 3.6 μm and 4.5 μm (Balmer break at $z \simeq 6.8$)
- F160W emission slightly extended along the N-S axis.

Band	Flux [μJy]	Band	Flux [μJy]
HSC g	<0.017	VIRCam K_s	0.67 $\pm$ 0.09
HSC r	<0.022	IRAC 3.6 μm	2.75 $\pm$ 0.10
HSC nb718	<0.056	IRAC 4.5 μm	2.77 $\pm$ 0.12
HSC i	<0.027	IRAC 5.8 μm	4.42 $\pm$ 1.04
ACS F814W	<0.027	IRAC 8.0 μm	<4.88
HSC nb816	<0.054	MIPS 24 μm	179.7 $\pm$ 6.4
HSC z	<0.030	PACS 100 μm	5100 $\pm$ 1370
HSC nb921	<0.051	PACS 160 μm	10130 $\pm$ 4260
HSC ib945	<0.062	SPIRE 250 μm	8830 $\pm$ 1120
HSC nb973	0.20 $\pm$ 0.06	SPIRE 350 μm	8090 $\pm$ 1600
HSC y	0.18 $\pm$ 0.04	SPIRE 500 μm	11130 $\pm$ 2220
VIRCam Y	0.22 $\pm$ 0.04	SCUBA-2 850 μm	6300 $\pm$ 1650
VIRCam J	0.37 $\pm$ 0.05	VLA 3 GHz	20.0 $\pm$ 2.6
WFC3 F160W	0.44 $\pm$ 0.12	VLA 1.4 GHz	96.3 $\pm$ 14.4
VIRCam H	0.56 $\pm$ 0.05	-	-

Observations of COS-87259

Radio at 1.4 and 3 GHz

- FWHM = $1.5''$ (1.4 GHz) and $0.75''$ (3 GHz)
- $\alpha_{3000}^{1400} = -2.06$ where $S_\nu \propto \nu^\alpha$ between 1.4 and 3 GHz, ultra steep (< -1.3)
- Not detected on 325 and 610 MHz, limiting $\alpha_{1400}^{325} > -1.11$ and $\alpha_{1400}^{610} > -0.85$ (high redshift $z > 4$ radio-loud galaxies)



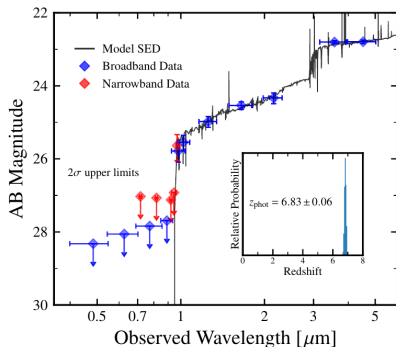
Observations of COS-87259

- Mid-Infrared, Far-Infrared, Sub-mm Observations
- X-ray: Not detected (a $z \simeq 7$ AGN)
- Spectroscopy between 7490 - 10010 Å: Not detected (an extremely dusty UV-bright $z \simeq 7$ galaxy)

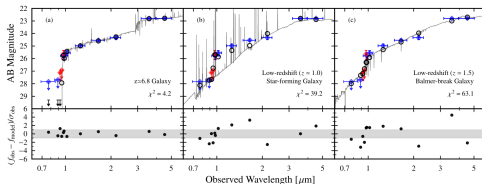
Physical Properties of COS-87259

Fitting the optical to near-IR photometry with Bayesian Analysis (BEAGLE)

- Ly α break at $0.95 \sim 1 \mu\text{m}$:
 $z = 6.83 \pm 0.06$
- **Red** rest-UV slope
 $\beta = -0.59$ where $F_\lambda \propto \lambda^\beta$ at
1500 Å: **strong dust
attenuation**
- Flux excess on $3.6 \mu\text{m}$ and
 $4.5 \mu\text{m}$: **relatively old
(~ 300 Myr) stellar
population producing a
strong Balmer break**



Physical Properties of COS-87259



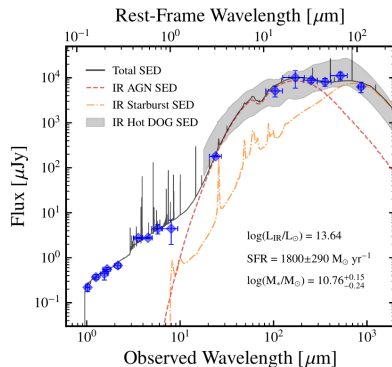
Possibility of Low-redshift Interpretation

- Dusty young galaxy with strong emission lines (star-forming)
- Old stellar population yielding a prominent Balmer break
- Low-redshift, star-forming galaxy but its $H\beta$ or $OIII$ emission lines are likely to be detected in spectrum
- FeLoBAL (iron low-ionization broad absorption line) AGN at $z \approx 2.4$, cannot easily explain extended morphology and non-detection of blue optical bands.
- $z \simeq 7$ galaxy gravitationally lensed by a dusty starburst galaxy: no evidence of lensing in HST images.

Physical Properties of COS-87259

Fitting the photometry with X-CIGALE assuming $z = 6.83$

- extremely red rest-UV slope and sub-mm flux density measurements are simultaneously reproduced by a **highly-obscured starburst**.
- the $24 - 160 \mu\text{m}$ measurements can be explained by hot dust emission from **an edge-on torus surrounding an AGN**, $\tau = 7.7 \pm 2.5$ at $9.7 \mu\text{m}$



Physical Properties of COS-87259

- The hyperluminous infrared emission and steep mid-infrared slope of COS-87259 suggests that this source may be a **higher-redshift analog of hot dust-obscured galaxies (hot DOGs)** which have very large stellar mass, heavily obscured AGN and recent, intense starbursts.
- X-ray non detection is consistent with this highly obscured AGN interpretation as in hot DOGs compared to Type 1 AGN.

Overdensity Around COS-87259

- **Observations:** Identified 4 $M_{UV} < -21$, $z \simeq 6.6 - 6.9$ neighbouring galaxies in an 18 arcsec^2 region, 11 times the average, suggesting COS-87259 may trace a highly overdense region at $z \simeq 6.8$, as is common for systems hosting radio-loud AGN at lower redshifts
- **Simulations:** 66% simulations by **MultiDark Planck 2** return the same result

Outlook

- At least 1/4 very massive, UV-bright galaxies in COSMOS exhibit **both AGN and highly-obscured star formation activity**
- **Very massive galaxies** contribute significantly to the cosmic **star formation** rate density at $z > 6$
- AGN are **fairly common** among the most massive, UV-luminous galaxies in the reionization-era
- AGN make a non-negligible contribution to **ionizing photon** at early times

Thank you for listening!